

6.	C	Work done = $120 \cos 37^\circ \times 5.0 = 479 = 480 \text{ W}$
7	B	Initially, when no force is applied to the piston, force on piston by air in container = force on piston by air in atmosphere = pressure x area = $(100 \times 10^3)(3.5 \times 10^{-3})$ = 350 N $\frac{(100 \times 10^3)(3.5 \times 10^{-3})(80 \times 10^{-3})}{T} = \frac{(\text{new pressure})(3.5 \times 10^{-3})(160 \times 10^{-3})}{0.5T}$ new pressure = 25 kPa new force on piston by air in container = pressure x area = $(25 \times 10^3)(3.5 \times 10^{-3})$ = 87.5 N

		For piston to remain stationary, Force on piston by air in container + F = Force on piston by air in atmosphere $87.5 + F = 350$ $F = 260 \text{ N (262.5)}$
9	C	Let X be the gas compressed isothermally (no change in temp). There is heat